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QCII Part 2: Application

2nd submission

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1 Electron Correlation

1.1 Multireference Methods

The dissociation of hydrofluoric acid (HF) was investigated, for bond distances between 0.5 and 4 Å using several computational methods in the def2-TZVP [20] basis. The dissociation curve was computed, using ORCA 5.0.3 [14, 13] at RHF, UHF, MP2, CCSD(T) levels of theory. In addition the multireference-method CASSCF [17] (complete active space self consistent field) was employed in two ways. Configurations were generated using either two or six active electrons in the same number of active orbitals. The resulting dissociation curves are depicted in Fig. 1 relative to the sum of the energies of the individual atoms. The true dissociation limit therefore lies at zero.

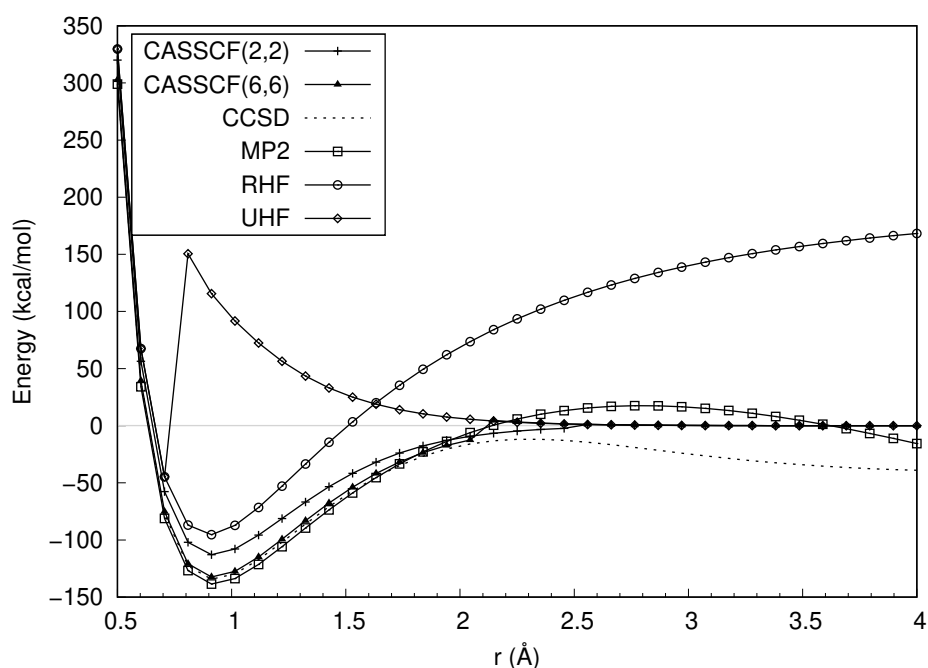


Figure 1: HF Dissociation for various methods

The RHF curve diverges towards higher values for larger distances as each pair of electrons is required to occupy the same spatial wavefunction (WF). In the case of HF this means that only ionic dissociation is possible, and as a result the energy of the dimer is too low with respect to the individual atoms. The incorrect description of the wavefunction in RHF can be alleviated by employing UHF. The unrestricted treatment yields the correct dissociation limit for distances beyond 2 Å but contains an unphysical cusp at 0.8 Å. The correct dissociation limit is obtained for a triplet state in the UHF WF which in turn does not relate to the UHF ground state at equilibrium distances, hence the instability point. For very small distances the UHF can be thought of as a pure singlet state. The dissociation curves of RHF and UHF

are hence identical below the instability point.

Both UHF and RHF yield too low binding energies around the equilibrium distance due to the lack of electron correlation. Post HF methods such as MP2 and CCSD(T) can be used to account for correlation effects in order to obtain accurate energies around the equilibrium. CCSD(T) is generally considered the “gold-standard” of computational chemistry and can be considered as the most accurate energy around the equilibrium here. However, for large distances the MP2 energy correction

$$E_0^{(2)} = \frac{1}{4} \sum_{ij,ab} \frac{|\langle ij||ab \rangle|^2}{\epsilon_i + \epsilon_j - \epsilon_a - \epsilon_b}, \quad (1)$$

approaches $-\infty$ as the denominator approaches zero. This is the case for large distances where the energy difference of occupied (i, j) and virtual orbitals (a, b) becomes smaller. As a result, the MP2 curve (squares) is lowered in energy for large distances. The same holds true, to a lesser extent, for CCSD(T) as triple excitations are computed perturbatively.

Both CASSCF calculations lead to the correct dissociation behaviour at large distances. CASSCF is a multireference-method which utilizes multiple ground state configurations for the computation of excited states. In order to limit the computational cost, the size of the active orbital space has to be reduced. CASSCF(2,2) is too high in energy around the equilibrium distance, but for large distances, generating configurations using only the two outermost electrons is sufficient. Including more orbitals and electrons in the active space leads to an accurate description around the equilibrium point. Both CASSCF calculations also include an instability point (2 Å for CASSCF(6,6) 2.5 Å for CASSCF(2,2)) similar to UHF. Contrary to UHF, the instability is much less pronounced and does not lead to an unphysical repulsive state.

1.2 Carbenes

The energy difference between singlet and triplet states ΔE_{S-T} was computed for the biradicals methylene and p-benzyne. H-C-H bond angles for singlet and triplet states in methylene were also computed based on the resulting geometries. Geometry optimizations and single point calculations were carried out using TURBOMOLE 7.6 with HF, MP2, TPSS, PW6B95 [21] and B3-LYP [4] in the def2-TZVP [20] basis. Additional single point calculations using CCSD(T) with MP2 geometries were also performed. Both ΔE_{S-T} and α_{HCH} are listed in Tab. 1 along with an experimental reference.

Table 1: Singlet Triplet energy splittings (kcal/mol) for p-benzyne and methylene along methylene HCH angles. All values were obtained using a def2-TZVP basis. CCSD(T) calculations were performed using MP2 geometries.

	ΔE_{S-T}		$\alpha_{HCH} (CH_2)$	
	p-Benzyne	Methylene	Singlet	Triplet
HF	80.5	28.4	103.7°	131.8°
MP2	-28.6	15.7	102.4°	133.0°
CCSD(T)	-3.5	10.6	-	-
B3LYP	13.4	11.9	101.9°	135.0°
TPSS	7.0	17.8	101.3°	135.4°
PW6B95	14.3	12.6	101.6°	134.8°
Experiment	-4.2	9.0	102.4°	135.5°

With the exception of CCSD(T) none of the employed methods yields chemical accuracy (i.e. 1 kcal/mol) for either system. CCSD(T) yields chemical accuracy for p-benzyne and deviates by 1.6 kcal/mol for methylene. HF vastly overestimates the energy splitting for the methylene biradical by 215% and is not even qualitatively correct for p-benzyne. A triplet needs to be calculated using UHF and therefore contains an artificial energy lowering due to spin contamination. HF therefore underestimates the electronic energy of the triplet w.r.t the singlet hence the large energy gaps favouring the triplet in both systems.

MP2 yields qualitatively correct energies for both molecules. However in both systems the value of ΔE_{S-T} is too large compared to experiment. This is especially pronounced for p-benzyne where MP2 overestimates the energy splitting by a factor of 7 (favouring the singlet). The large error is caused by a small HOMO-LUMO gap in singlet p-benzyne which leads to a large overcorrection of the correlation energy (see Eq. (1)).

In the case of methylene the DFT functionals follow Perdew’s *Jacob’s Ladder* scheme. The meta-GGA TPSS performs significantly worse than the hybrids PW6B95 and B3LYP, with the latter yielding the best energy among the DFT functionals. Both hybrid functionals lie between MP2 and CCSD(T) in terms of accuracy for methylene. For p-benzyne, the self interaction error (SIE) in DFT leads to the triplet being more stable than the singlet for all molecules.

Regarding the methylene HCH angles in Tab. 1, all methods reproduce the experimental values with decent accuracy. The worst results are achieved with HF with deviations of 1.3° and 3.7° for the singlet and triplet respectively. MP2 yields the correct singlet geometry although the triplet angle is 2.5° lower than the reference. Contrary to *Jacob’s ladder*, TPSS yields the most accurate angle for the triplet with a deviation of 0.1°. Among the DFT methods, B3LYP reproduces the singlet angle the best with a deviation of 0.5°.

2 Basis Set Convergence

The basis set convergence behaviour for the formic acid dimer, depicted in Fig. 2, was investigated at HF, MP2 and TPSS-D3 levels of theory. Dimerization energies were computed using the aforementioned methods in cc-pVXZ [6] basis sets along with their augmented-counterparts. Cardinal numbers ζ from 2-4 were used. All single point calculations used geometries, optimized at TPSS-D3/def2-TZVP level. The resulting energies are depicted in Fig. 3. The dimerization energies in Fig. 3 are generally prone to two types of errors. On

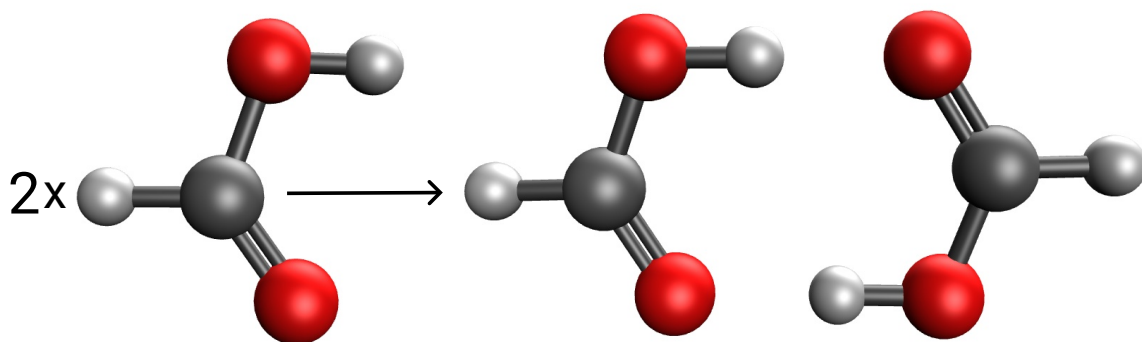


Figure 2: Dimerization of the formic acid dimer. Geometries were optimized using TPSS-D3/def2-TZVP.

one hand the basis set incompleteness error (BSIE) is caused by the difference between the employed basis sets and the complete basis-set (CBS) limit. However, this error is not specific to dimerization and always present unless one extrapolates towards the CBS-limit. It should be noted that the convergence behaviour with regards to the basis set depends on the employed method. SCF energies (i.e. HF/DFT) converge exponentially w.r.t basis set size whereas correlation energies converge with inverse cubic behaviour.

On the other hand dimerization energies are particularly prone to the basis set superposition error (BSSE). When computing dimers the basis functions of the monomers overlap and compensate for the incompleteness of the basis set. As a result the energy of the dimer is artificially lowered in comparison to the monomer. Nonetheless, this error is not inherent to dimer calculations, as it also occurs intramolecularly. Only calculations using a minimal basis do not suffer from BSSE. Both BSSE and BSIE are reduced by including a larger basis set. In Fig. 3 the effect of BSSE becomes apparent. With the exception of MP2/aug-cc-pVXZ all dimerization energies become smaller with increasing cardinal number. This effect is significantly more pronounced in the cc-pVXZ basis. The augmented basis sets contain additional diffuse functions which lowers the BSIE and in return also the BSSE. Both errors can, in principle be accounted for, the BSIE by exploiting the convergence behaviour men-

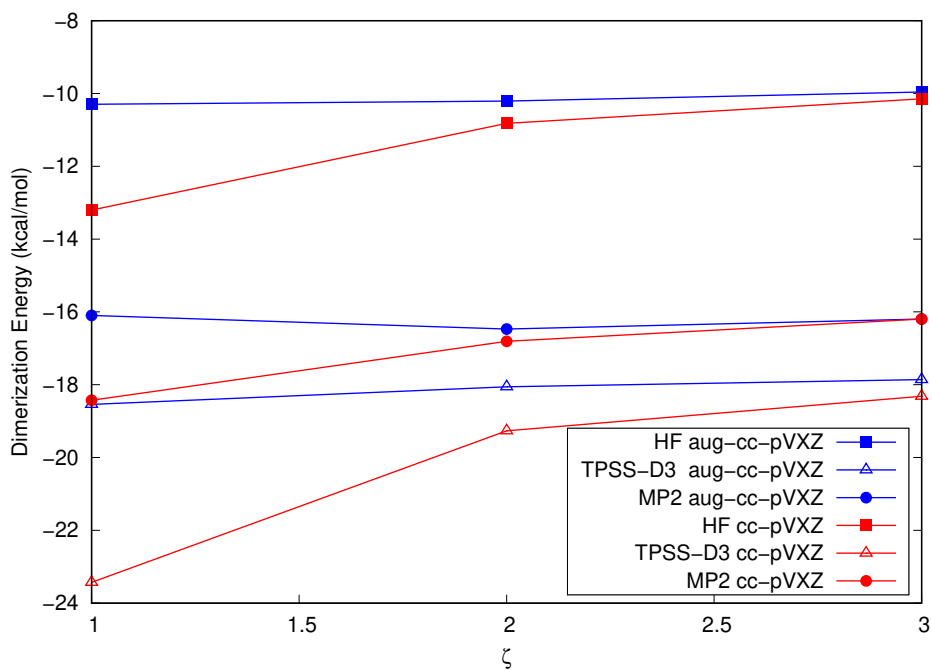


Figure 3: Convergence behaviour for the formic acid dimer aug-cc-pVXZ and cc-pVXZ basis sets at HF, MP2 and TPSS-D3 levels of theory.

tioned above. In order to alleviate BSSE the *Boys-Bernardi-Counter-Poise* correction can be calculated, which tends to overcorrect the BSSE.

3 Thermochemistry

The reaction enthalpies ΔH of two industrially important reactions, namely the Haber-Bosch process and steam reforming of methane, were computed.



Geometries of all products and reactants were optimized using TPSS-D3/def2-TZVP. For comparison, geometries of molecules involved in the Haber-Bosch process were also optimized on the MP2/def2-TZVP level and the RMSD to TPSS-D3 geometries were computed. The RMSD values are displayed in Tab. 2.

Table 2: RMSD between TPSS-D3 and MP2 geometries optimized in a def2-TZVP basis for all molecules involved in the Haber-Bosch process.

	RMSD (Å)
N ₂	$6.33 \cdot 10^{-3}$
H ₂	$5.96 \cdot 10^{-3}$
NH ₃	$3.07 \cdot 10^{-3}$

As evident by the RMSD values in Tab. 2 optimizing geometries at the MP2 level only makes a marginal difference compared to TPSS-D3. The choice of computational method for geometry optimizations is thus governed mainly by its computational cost. Single point calculations for all molecules were hence carried out using TPSS-D3 geometries. Thermal corrections to the enthalpy were computed and single point calculations were carried out at B3LYP-D3/def2-TZVP, MP2/def2-TZVP, B2PLYP-D3/def2-QZVP and CCSD(T)/def2-QZVP levels of theory. Using the single-point energies and thermal corrections, reaction enthalpies were computed for both reactions at 298 K. The resulting enthalpies are displayed in Tab. 3.

Table 3: Reaction enthalpies for the Haber-Bosch process and steam-reforming of methane. Geometries were optimized on the TPSS-D3/def2-TZVP level. B2-PLYP-D3 and CCSD(T) energies were computed in a def2-QZVP basis, def2-TZVP was used for all other methods.

	MP2	TPSS-D3	B3-LYP-D3	B2-PLYP-D3	CCSD(T)
Haber-Bosch	-12.1	-12.7	-21.0	-23.9	-22.3
	Exp: -22.5 kcal/mol				[16]

Table 3: Reaction enthalpies for the Haber-Bosch process and steam-reforming of methane. Geometries were optimized on the TPSS-D3/def2-TZVP level. B2-PLYP-D3 and CCSD(T) energies were computed in a def2-QZVP basis, def2-TZVP was used for all other methods.

	MP2	TPSS-D3	B3-LYP-D3	B2-PLYP-D3	CCSD(T)
Steam-Reforming	43.3	42.3	47.31	50.5	49.0
	Exp: 49.3 kcal/mol				[16]

As expected CCSD(T) yields the most accurate enthalpies of all the employed method, achieving sub chemical accuracy (1 kcal/mol) for both reactions. The least accurate values are produced by MP2 and TPSS-D3 with both methods yielding deviations of around 10 kcal/mol for the Haber-Bosch process, and 6/7 kcal/mol for steam-reforming. In general the accuracy of the DFT functionals follows *Jacob's Ladder*. The meta-GGA TPSS-D3 is less accurate than the hybrid B3-LYP-D3 which in return is less accurate than the double-hybrid B2-PLYP-D3. It should, however, be noted that in the latter case a larger basis set was used, which makes a direct comparison more difficult.

4 Kinetics

The kinetic isotope effect of the hydrogen transfer from methane to the hydroxyl radical was investigated by employing a transition state search as implemented in TURBOMOLE 7.6 [19] at the B3-LYP-D3 [4]/def2-TZVP level. Starting structures were manually constructed with C–H and O–H distances in mind as depicted in Fig. 4. The transition state search was performed for both CH₄–OH and CD₄–OH. The electronic energies of all reactants were also computed in addition to thermal corrections in order to obtain activation enthalpies $\Delta H^\ddagger$.

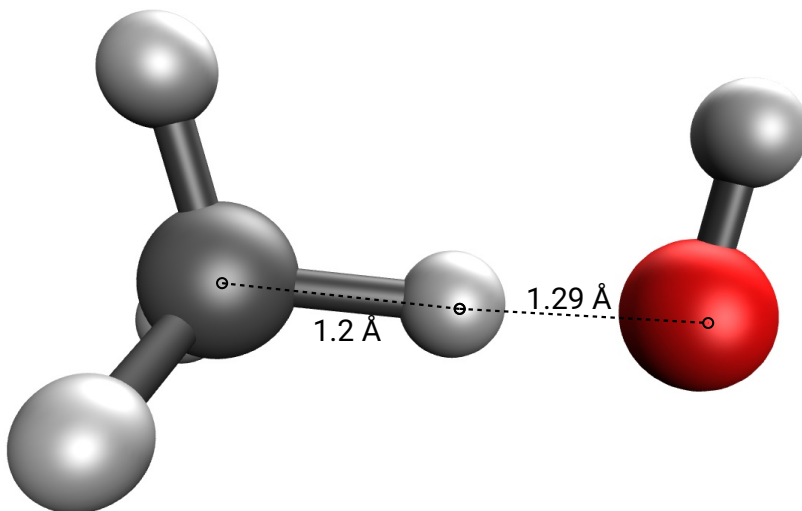


Figure 4: Geometry of the transition state.

Table 4: Electronic energies (E_h) along with zero-point vibrational energies and thermal corrections (298 K) for reactants and transition states.

	CH ₄	CD ₄	OH	CH ₄ –OH	CD ₄ –OH
E_{el} (E_h)	-40.504	-40.504	-75.735	-116.234	-116.234
ZPVE (kcal/mol)	27.962	20.502	5.255	31.719	25.188
Thermal correction (kcal/mol)	2.393	2.468	2.073	3.504	3.758

The energies in Tab. 4 amount to

$$\Delta H_H^\ddagger = 0.782 \text{ kcal/mol}$$

$$\Delta H_D^\ddagger = 1.890 \text{ kcal/mol},$$

with the subscripts H and D referring to regular and deuteriated methane respectively. As evident by the electronic energies in Tab. 3 the difference in $\Delta H^\ddagger$ stems solely from the different ZPVE and thermal corrections. The electronic energies are invariant to the mass of the nuclei in the molecule (unless one considers relativistic effects) and are hence identical for both isotopologues. Conversely the vibrational, rotational and translational energies which make up the thermal corrections are directly dependent on the mass of the nuclei. For instance, the vibrational frequencies

$$\nu = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}}, \quad (4)$$

depend on the reduced mass μ :

$$\mu = \frac{m_1 \cdot m_2}{m_1 + m_2}. \quad (5)$$

The relative reaction rates can then be determined from $\Delta H^\ddagger$ according to:

$$\frac{k_H}{k_D} = e^{-\frac{\Delta H_H^\ddagger - \Delta H_D^\ddagger}{RT}} = 7.09 \quad (6)$$

According to the presented results, the hydrogen/deuterium transfer from methane to the hydroxy radical proceeds seven times faster with regular methane w.r.t. its deuteriated isotopologues. The rather high value is likely influenced by the large number of vibrational modes in methane ($3 \cdot 5 - 6 = 9$).

5 Solvation

The effect of solvation on an S_N2 reaction of Chloromethane with Fluoride was investigated by calculating the potential energy curve between C – F distances of 2.25 Å – 8 Å. The potential energy curve was computed using the implicit solvation model COSMO [11] for methanol with a dielectric constant $\epsilon = 32.0$. Additionally the same potential energy curve was computed in the absence of a solvent. Both calculations were carried out using TURBOMOLE 7.6 [19] using the PW6B95 [21] functional in a def2-TZVP [20] basis, employing the D3 dispersion correction with Becke-Johnson damping [7]. At every point in the potential curve, the C – F distance was kept fixed, while the geometry of the Cl- and H-atoms were optimized. The resulting potential curve is depicted in Fig. 5, a ball and stick model of the transition state in methanol in Fig. 6.

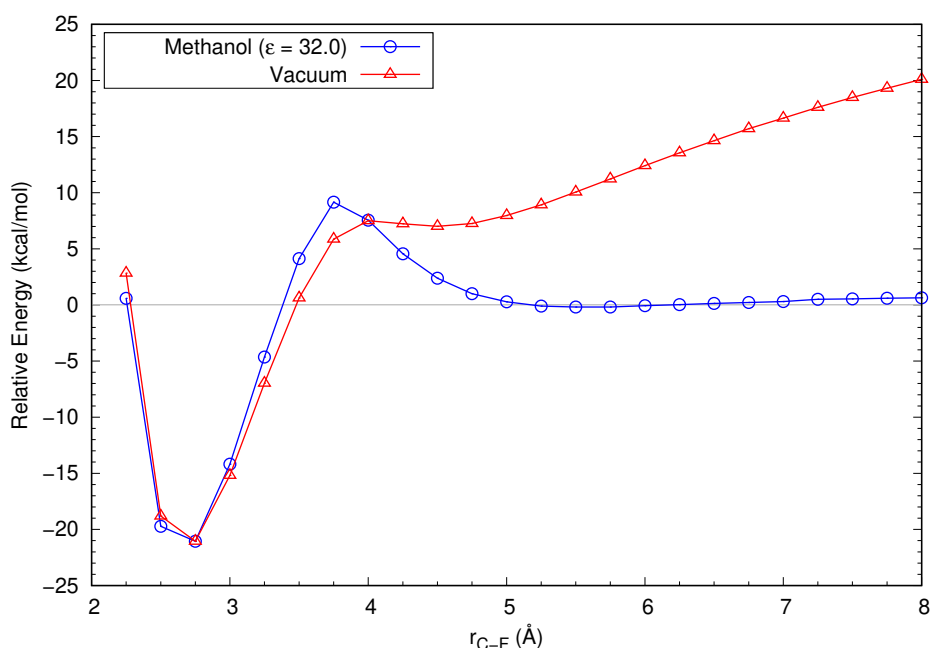


Figure 5: Potential energy curve for the S_N2 reaction of Chloromethane and Fluoride. Blue circles: In presence of methanol as an implicit solvent. Red triangles: In vacuum. The blue curve has been adjusted to the dissociation limit and the red curve was shifted such that both minima align.

The potential energy curves in Fig. 5 exhibit maxima corresponding to the transition state at 3.75 Å and 4 Å respectively. The different C – F distances in the transition state are a consequence of the charge shielding effect of the solvent. The geometry of the transition state in methanol is depicted in Fig. 6. The activation barriers for the reactions can be estimated based on the potential curves:

$$\Delta E_{solv}^{\ddagger} = 9.2 \text{ kcal/mol mol}$$

$$\Delta E_{gas}^{\ddagger} = 0.5 \text{ kcal/mol mol}$$

The reaction barrier in solvent is almost 20 times as large as the respective S_N2 reaction in vacuum. However comparing activation barriers is not really possible due to the vastly different behaviour of the potential curves at long ranges. In vacuum the fluoride anion is unstable as there is no opposing charge to compensate for. As the fluoride anion approaches the chloromethane molecule its charge becomes stabilized by the dipole of chloromethane. In contrast, when the reaction proceeds in a protic solvent such as methanol, the charge of the anion is compensated for and the potential curve converges towards the dissociation limit. The reference states for the reaction barriers are thus vastly different which should be accounted for when interpreting the activation barriers.

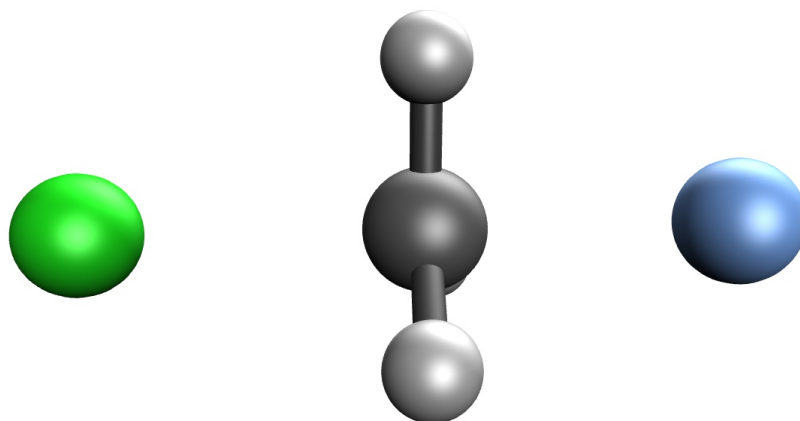


Figure 6: Transition state of the S_N2 reaction of Chloromethane and Fluoride (blue) in methanol. The geometry corresponds to the point of the blue curve in Fig. 5 at 3.75 Å.

The continuum solvation model COSMO calculates solvent interactions by modelling the solvent as a dielectric continuum outside of a molecular cavity, representing the solute. The charge density on the surface of the molecular cavity polarizes the dielectric continuum which in turn stabilizes the charge on the surface of the molecular cavity. As a result, the (attractive) interactions from electrostatics, polarization, and to a lesser extent, dispersion and charge-transfer are screened by the solvent [12]. When the molecular cavities approach each other they eventually unite, thus drastically decreasing the screening effects introduced by the solvent. Upon decreasing the bond distances the molecular cavities of both fragments start to unite which drastically decreases these screening effects.

6 Activation Energies

The reaction paths for the Claisen rearrangement of allyl-vinyl ether and the dimerisation of cyclopentadiene were investigated using the double ended growing string method (GSM) at the GFN2-xTB [5] level. Geometries of the reactants and products were optimised using GFN2-xTB. Then, a transition state search with the double ended GSM was carried out. 25 points along the reaction path were computed for the Claisen-rearrangement and 20 points were used for the Diels-Alder reaction. Following the transition state search, activation energies for both reactions were also determined by performing single-point DFT calculations on the reactants and transition states at the PW6B95 [21]/def2-TZVP [20] level. The energy diagram of the reaction paths are displayed in Fig. 8 (Claisen-rearrangement) and Fig. 9.

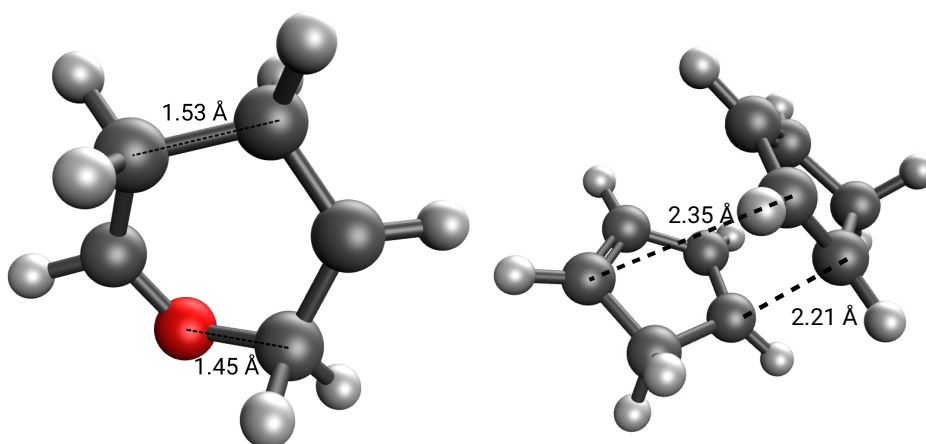


Figure 7: Transition states for the claisen rearrangement and Diels-Alder reaction. Important bond distances were highlighted

The reaction path of the Claisen-rearrangement proceeds via a cyclic transition state with a lengthened C–O bond length of 1.45 Å hinting at the subsequent bond breakage. The activation barrier for the reaction can be estimated using Fig. 8 yielding

$$\Delta E^{\ddagger} = 39.5 \text{ kcal/mol},$$

using the single point PW6B95 yields an activation energy of

$$\Delta E^{\ddagger} = 65.2 \text{ kcal/mol}.$$

Both determined activation energies seem rather high, especially the DFT value. This may indicate a poor transition state found by the double ended GSM, emphasizing the liabilities of the method. The high energy difference in the PW6B95 calculation can be seen as a consequence of the poor transition state. Experimentally the activation energy has been determined as 30.6 kcal/mol [18] in the gas-phase and 25.4 kcal/mol when solvated in dibutyl ether (at). According to the Boltzmann-factor a difference in activation energy of around

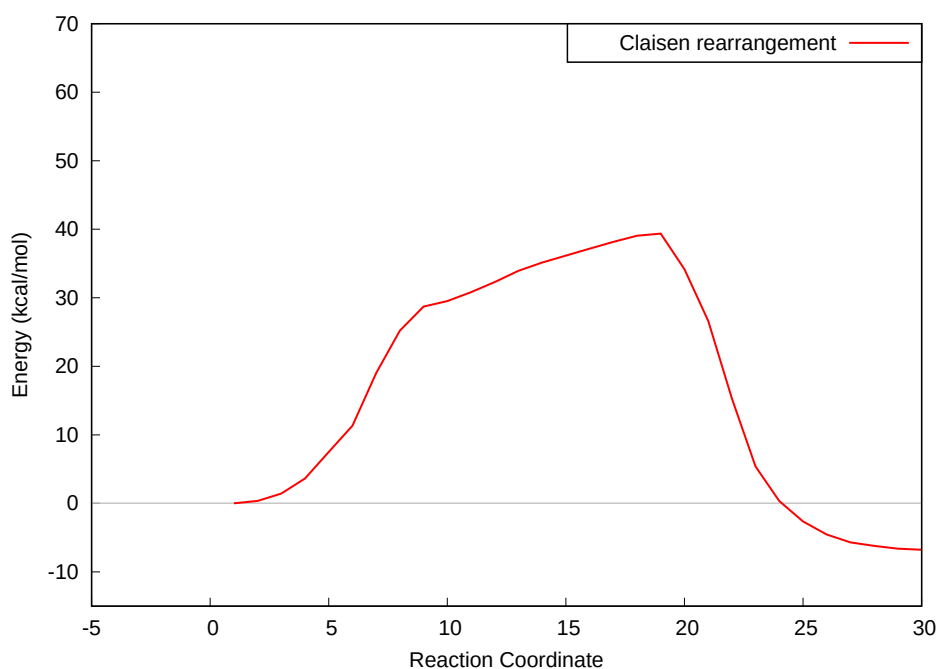


Figure 8: Reaction path for the Claisen rearrangement of allyl-vinyl ether. The energy of the reactand was set to zero.

10 kcal/mol amounts to a reaction rate which is million times faster at 480 °C (the temperature reported by [18]).

The reaction path of the Diels-Alder dimerisation of cyclopentadiene yields an activation energy of

$$\Delta E^{\ddagger} = 10.5 \text{ kcal/mol},$$

and using the single point PW6B95:

$$\Delta E^{\ddagger} = 18.3 \text{ kcal/mol}.$$

For both reactions, the hybrid-DFT calculation yields a significantly larger ($\approx 50\%$ for both reactions) barrier-height w.r.t. GFN2-xTB. The latter can be described as GGA-like and, in general, GGA and meta-GGA functionals underestimate barrier-heights [9]. The opposite holds true for HF calculations, which is why larger reaction barriers are obtained for hybrid-DFT. One should typically expect the PW6B95 reaction barrier to be more accurate although this does not apply here, probably due to wrong transition-state geometries.

The reliability of the transition state search may be improved by utilizing a higher level DFT calculation for the GSM. However, this would also increase the computational cost drastically as geometries need to be optimised on every step of the reaction paths. In general the GSM method is particularly sensitive towards the input geometries. One should also note that only a single transition state is found using this method. Whereas multiple transition states may be possible for a given reaction at the same time.

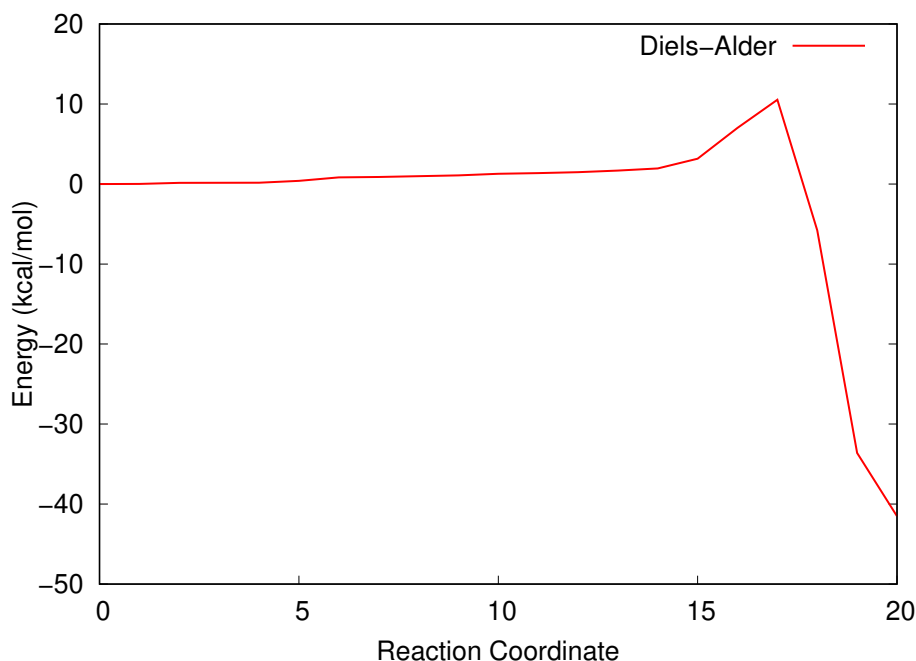


Figure 9: Reaction path, Diels-Alder

7 Noncovalent Interactions

The ‘weak’ interactions between methane and the noble gases argon and krypton were investigated for distances between 4.5 Å and 10 Å. The dissociation curves were computed at BLYP/def2-TZVP, BLYP-D3/def2-QZVP and MP2/def2-QZVP levels of theory. The D3 dispersion correction was applied with Becke-Johnson damping [**dispersion**, 7]. The resulting curves are plotted in Fig. 10. The importance of accounting for dispersion when employing DFT methods becomes apparent in Fig. 10. The uncorrected DFT functional (B3-LYP) yields a repulsive potential across all distances. This result is unphysical and not reproduced in either the MP2 or the dispersion corrected curve.

Dispersion is an attractive interaction between molecules caused by the transition dipole moment induced by coupled single excitations. Mean field theories such as KS-DFT do not account for coupled single excitations and thus neglect dispersion. As a consequence including Fock-exchange in a KS-DFT (such as in a hybrid) calculation does not alleviate this issue. Post-HF methods such as MP2 inherently include the aforementioned excitations in the theory which eliminates the need for an additional dispersion correction.

In fact, the dissociation curves of B3-LYP-D3 and MP2 agree with one another almost perfectly. Both yield a bound minimum around 5.9 Å for Ar–HCH₃ and 6.1 Å for Kr–HCH₃ with binding energies of roughly 0.2-0.3 kcal/mol. The binding energy for the Kr–HCH₃ dimer is slightly larger due to the higher polarizability of Kr w.r.t Ar. Due to the long-range nature of dispersion interactions, utilizing a large basis-set with diffuse basis-functions is beneficial for an accurate assessment, hence the def2-QZVP basis set.

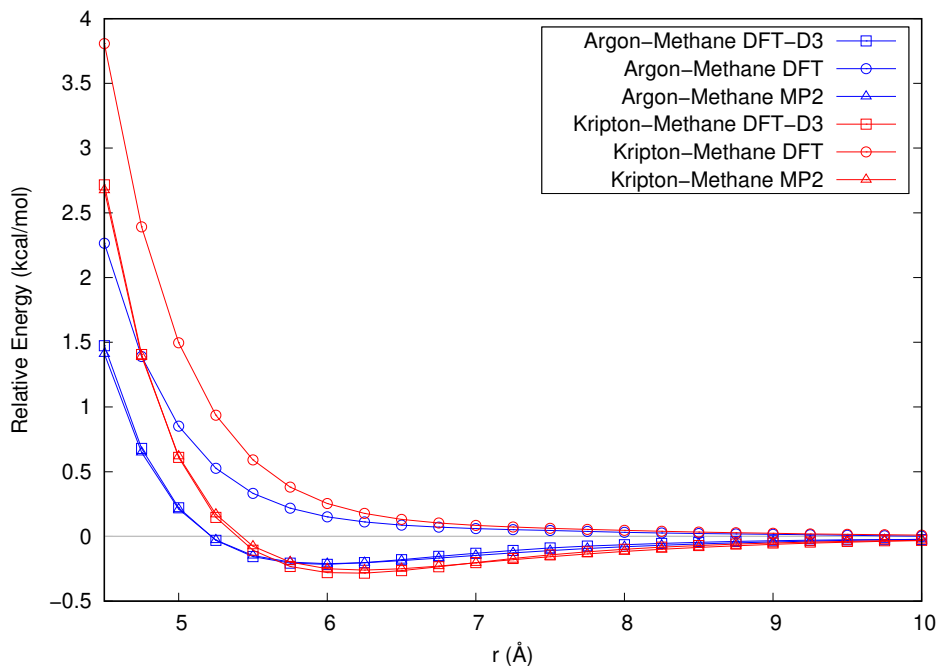


Figure 10: Dissociation Curves for Noble Gas Methane Dimers

8 Spectroscopy

8.1 IR-Spectrum

A theoretical IR-Spectrum of 1,4-Benzoquinone was calculated at HF-D3/def2-SVP and TPSS-D3/def2-SVP levels of theory. The geometry was optimised at the respective level using TURBOMOLE 7.6 [19]. Subsequently the vibrational modes were generated using the subprogram aoforce. The frequencies were then analyzed so as to confirm that the calculated frequencies are indeed IR-active. The vibrational spectra for HF-D3 and TPSS-D3 are depicted in Fig. 11 along an experimental reference [1]. The intensities were normalized such that the most intense peak of all spectra have the same intensity. Selected vibrational frequencies of the theoretical spectra were then assigned to a corresponding mode in the experimental spectrum. A scaling factor was determined for each of the selected frequencies according to:

$$f_{\text{scal}} = \frac{\tilde{\nu}_{\text{exp}}}{\tilde{\nu}_{\text{calc}}}, \quad (7)$$

with $\tilde{\nu}_{\text{exp/calc}}$ referring to the wavenumbers of the experimental and calculated spectra respectively. The scaling factors of all considered frequencies are listed in Tab. 5.

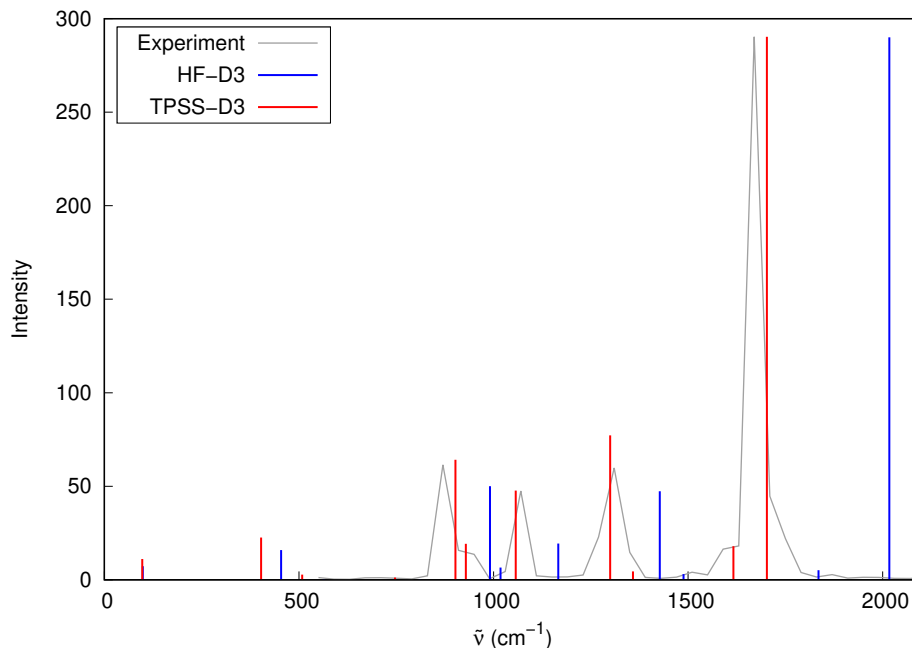


Figure 11: IR Spectrum of 1,4-Benzoquinone calculated using HF-D3 (blue) and TPSS-D3 (red). The experimental reference was obtained from [1].

Table 5: Wavenumbers of IR-modes (cm^{-1}) calculated at the HF-D3/def2-SVP and TPSS-D3/def2-SVP level along an experimental reference [1]. Each theoretical frequency has been assigned a scaling factor (see Eq. (7))

Exp		HF-D3			TPSS-D3		
$\tilde{\nu}$	I	$\tilde{\nu}$	f_{scal}	I_{rel}	$\tilde{\nu}$	f_{scal}	I_{rel}
1670.0	1	2017.0	0.83	1	1702.4	0.98	1
1590.0	0.06	1835.13	0.87	0.02	1616	0.98	0.06
1310.0	0.21	1427.2	0.92	0.16	1299.7	0.99	0.26
1070.0	0.16	1166.53	0.92	0.07	1057.1	1.01	0.16
870	0.21	990.97	0.88	0.17	902.3	0.96	0.22

The scaling factors of the considered frequencies average out to $f_{\text{scal}} = 0.88$ for HF-D3 and $f_{\text{scal}} = 0.98$ for TPSS-D3. Overall HF-D3 significantly overestimates the actual wavenumbers as it underestimates the bond-strength. The relative intensities also show noteworthy deviations from experiment. On the other hand, TPSS-D3 is almost spot on regarding both frequencies and relative intensities. TPSS-D3 is a meta-GGA meaning it includes the kinetic energy of the electrons in its exchange correlation potential. It therefore yields good approximations for force constants and thus vibrational frequencies.

The computation of IR-frequencies requires the calculation of the Hessian-matrix which is

the second derivative of the potential energy surface. The frequencies were thus only calculated in a minimal basis-set with efficient computational methods. Improving upon the presented results by using more accurate methods in a larger basis set (e.g. hybrid functionals) may be possible but was not feasible in the scope of this practical course.

8.2 The Color of Indigo

The UV/Vis transition of the Indigo molecule was calculated using time-dependent TD-HF and TD-DFT methods. The geometry of the indigo molecule was optimised at the TPSS-D3/def2-SVP level of theory. Consequently the time-dependent calculations were carried out, namely HF, PBE [15] and PBE0 [2] in a def2-SVP [20] basis. The wavelengths of the optical transition and the associated colour are noted in Tab. 6.

Table 6: Optical transition the indigo molecule calculated using a variety of methods in def2-SVP basis.

	HF	PBE	PBE0	Exp
λ (nm)	426.1	617.3	532.9	610
Colour	yellow/orange	blue	violet	blue

PBE reproduces the value with good accuracy deviating from the reference by merely 7 nm. Both of the other methods vastly overestimate the HOMO-LUMO gap and, in return, underestimate the absorption frequency by 77 nm (PBE0) and 184 nm (HF) respectively. Most noteworthy, the hybrid PBE0 functional performs significantly worse than the GGA PBE. The classical expectation of *Jacob's Ladder* is not reproduced here. The bad performance of HF may be explained by the high energy upshift of its virtual orbitals [3].

8.3 NMR

The ^{13}C chemical NMR-shifts of the organic molecules A-G (see Fig. 12) were calculated with TMS as a reference. Geometries were first optimised at the PBEh-3c [8] level of theory using the corresponding module in ORCA 5.0.3 [14, 13]. NMR chemical shieldings were subsequently calculated at the PBE [15]/pcSseg-2 [10] level of theory. The determined chemical shifts are listed in Tab. 7.

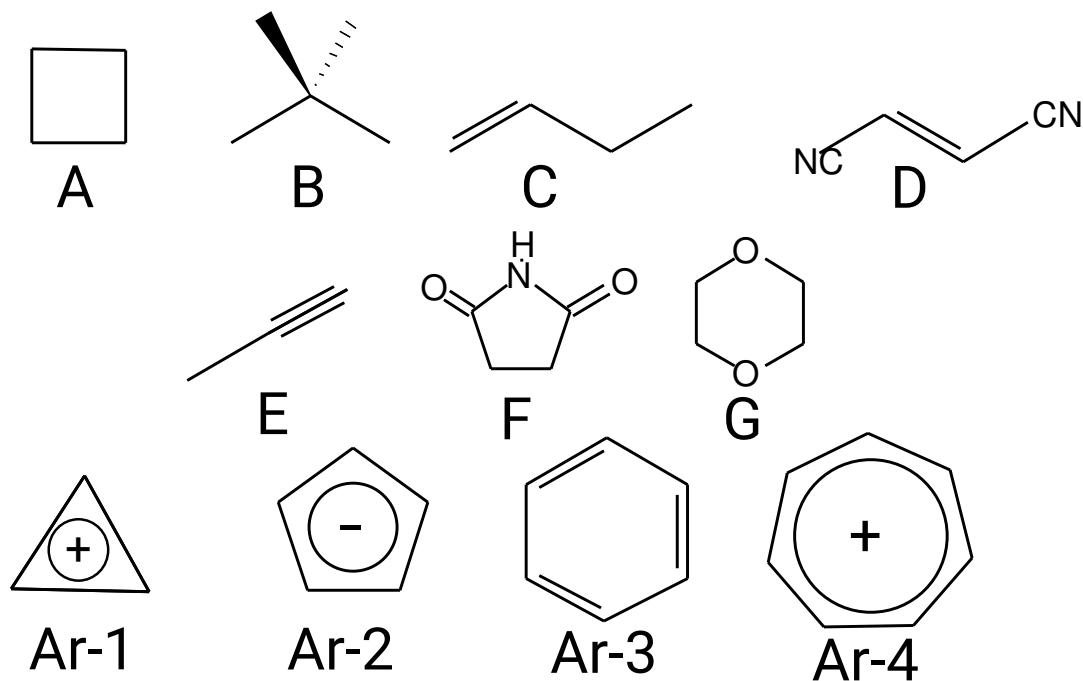


Figure 12: Lewis structures of organic compounds A-G, and Arenes Ar-1 - Ar-4, used in the NMR calculations.

Table 7: ^{13}C chemical shifts (δ) for several organic molecules in reference to TMS. Geometries were optimized using PBEh-3c and shielding tensors were calculated at the PBE/pcSseg-2 level of theory.

		δ (ppm)			
A	Calc			27.0	
	Exp.			22.3	
B	Calc	33.4			33.9
	Exp.	27.4			31.4
C	Calc	121.8	155.5	36.4	17.35
	Exp.	113.2	140.5	27.4	13.4

Table 7: ^{13}C chemical shifts (δ) for several organic molecules in reference to TMS. Geometries were optimized using PBEh-3c and shielding tensors were calculated at the PBE/pcSseg-2 level of theory.

		δ (ppm)		
D	Calc	124.0		128.0
	Exp.	120.4		123.6
E	Calc	4.04	71.2	85.4
	Exp.	1.9	66.9	79.2
F	Calc	33.9		182.2
	Exp.	30.3		183.6
G	Calc		71.4	
	Exp.		67.6	

The mean deviation (MD) and mean absolute deviation (MAD) from the experimental references were then calculated:

$$\text{MD} = \frac{1}{N} \sum_{i=1}^N (\delta_{\text{calc}} - \delta_{\text{exp}}) = 5.09 \text{ ppm} \quad (8)$$

$$\text{MAD} = \frac{1}{N} \sum_{i=1}^N (|\delta_{\text{calc}} - \delta_{\text{exp}}|) = 5.28 \text{ ppm.} \quad (9)$$

The similar values of MD and MAD reflect a general trend in the NMR calculations: In the vast majority of cases the theoretical chemical shifts overestimate the experimental ones. The computed values are thus somewhat more reliable than the MAD may suggest, as a constant offset can be accounted for. Large deviations from experiment were generally only encountered for large chemical shifts (e.g. alkene C-atoms for molecule C).

The correlation between formal π electron density and chemical shift was also investigated for the aromatic molecules Ar-1 through Ar-4. The formal π electron density ρ_{π} was determined as follows:

$$\rho_{\pi} = \frac{n_{\pi\text{el}}}{n_{\pi\text{at}}} \quad (10)$$

^{13}C chemical NMR-shifts were determined completely analogous to the molecules A – G. The resulting data is plotted in Fig. 13. The relation between ρ_{π} and the chemical shift of the Arene can be approximated rather well using a linear fit considering the crude approximation made for ρ_{π} . This results from the magnetic moment created by the aromatic ring current which results in increased shielding. As the chemical shift represents the difference in shielding

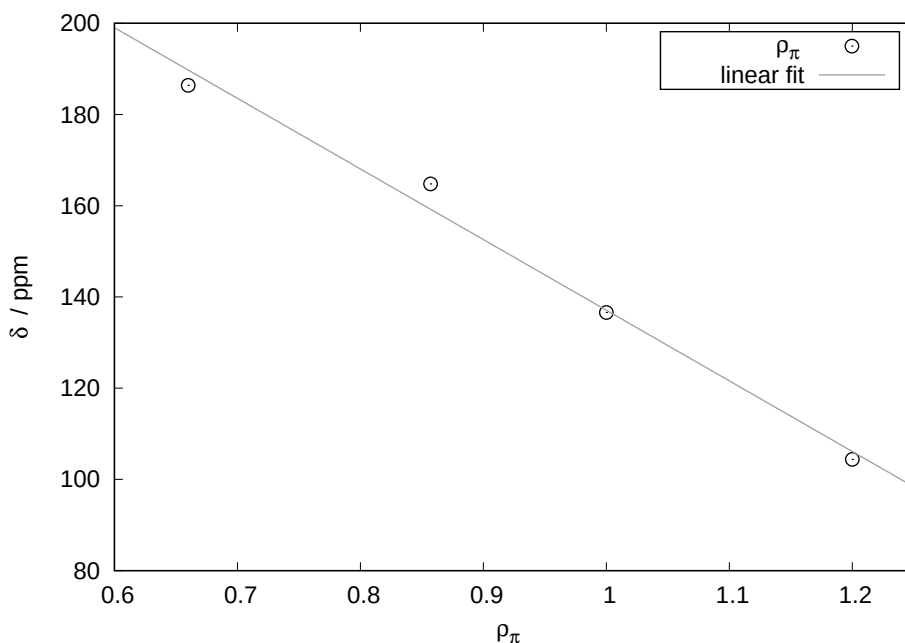


Figure 13: δ plotted against the formal π -electron density ρ_π including linear fit.

with respect to TMS the δ values drop with increasing ρ_π .

The effect of solvation on chemical shifts was also investigated on the example of acetone in water. Geometry optimizations and shielding tensor calculations were again performed as previously described with the addition of the Conductor like Polarisation Model (CPCM). The solvent model was included in the NMR calculation and/or in the geometry optimization step. Overall all possible combinations were investigated so as to determine during which step the inclusion of a solvation model has the largest impact. The shielding tensors were calculated for ^{13}C and ^{17}O with TMS and water as a reference. The resulting chemical shifts are listed in Tab. 8.

Table 8: chemical shieldings relative to TMS (^{13}C) and water (^{17}O) (δ in ppm). Reference data was optimized and calculated in gas-phase.

Geom	NMR	^{13}C		^{17}O	
		δ	$\Delta\delta_{\text{gas/gas}}$	δ	$\Delta\delta_{\text{gas/gas}}$
Gas	Gas	214.95	0.00	667.13	0.00
Gas	Solvent	226.29	11.35	616.54	-50.59
Solvent	Gas	216.09	1.15	674.78	7.65
Solvent	Solvent	227.35	12.40	620.85	-46.28

Upon the inclusion of a solvation model the ^{13}C chemical shifts increase across the board

regardless of the step in which they are included or which nucleus is considered. In experiment the ^{13}C -chemical shift for the solvated molecule differs by 18.9 ppm from the gas-phase. The presented calculation yields a lower value (12.4 ppm). Regarding the ^{17}O shifts, the experimental reference shows a difference of -75.5 ppm for gas/solvent. Again, the value could not be reproduced exactly but is nonetheless qualitatively correct. In both cases, performing the NMR calculation in solvent had a much bigger impact w.r.t the geometry optimization.

Lastly the ^1H chemical shifts of the methine proton for the isomers of $[3^{4,10}][7]$ metacyclophane were considered. Again, the calculations were carried out as previously described.

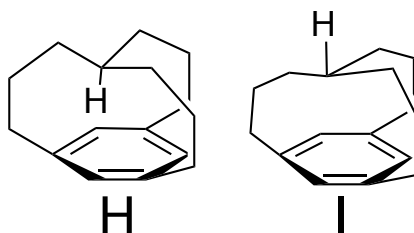


Figure 14: Lewis structures of in- (H) and out-(I)[$3^{4,10}$][7]metacyclophane

The metacyclophane isomers are distinct in their position of the methine proton which points either inwards (“in”) or outwards (“out”). As a consequence the influence of the aromatic ring current differs for the two isomers.

in: $\delta(^1\text{H})$: -4.04

out: $\delta(^1\text{H})$: 2.08

As one would expect, the methine proton pointing towards the aromatic ring current experiences a much larger shielding which results in the negative δ value relative to TMS.

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